

MATH-242
(Spring2026)
Homework 4

Due date: 12 March 2026

1 Instructions

1. Use this assignment format to be uploaded on MOODLE.
2. Show all steps of your work clearly.
3. No partial credit for unjustified answers.
4. Extra pages of work can be attached before uploading.
5. After the due date 20% marks will be deducted.
6. If you are going to use MATLAB for the first time, please get acquainted with basic commands and codes from the tutorial provided at <https://www.mathworks.com> or from any other suitable site.

Question 1. (22 points)

Find the solution to the following system of difference equations using the given initial conditions:

$$x_{n+1} - 7x_n - 10y_n = 4$$

$$y_{n+1} - x_n - 4y_n = 2$$

with $x_0 = 4$ and $y_0 = 1$. (Solve for x_n and y_n)

Question 2. (23 points)

Find the general solution of the differential equation using the method of variation of parameters.

a) $y'' - y = e^x$ (11 pts)

b) $y'' - 3y' + 2y = e^x$ (12 pts)

Question 3. (23 points)

Find the general solution for $x(t)$ in the following systems of differential equations.

a) $x' + 3y = x$ (11 pts)
 $x' + y + 2x = 0$

b) $x' - 2x + 4y = t$ (12 pts)
 $x' + x + y = 0$

Question 4. (22 points)

Find the solution to the following difference equations using the method of undetermined coefficients.

a) $y_{n+2} - 8y_{n+1} + 15y_n = 3^n$; $y_0 = 1, y_1 = 10$ (11 pts)

b) $y_{n+2} - 2y_{n+1} - y_n = 3 \cos n$ (11 pts)

Question 5. (10 points)

Consider the following differential equation with an initial condition:

$$y' = y + 5x - 2x^2, \quad y(0) = 1.$$

Solve the differential equation using the fourth order Runge-Kutta method in MATLAB. Then, use MATLAB's `ode45` function to solve the same differential equation. Plot both solutions over a suitable interval in the same figure.